

PEYTON RAMSEY

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EDUCATION

University of New Hampshire, Durham, NH | MS in Business Analytics (MSBA) (GPA: 3.83) | August 2025 – May 2026

Graduate Research Assistant: Data Science & AI

- Developed “AI Tools and Apps for Business” course teaching 20 MSBA students production ML deployment; created slides on Docker, Hugging Face, LLMs, Git, and responsible AI practices with hands-on labs and interactive demos.
- Modernized 12+ modules in Predictive Analytics and Big Data; migrated course materials from RMD to Quarto.

University of New Hampshire, Durham, NH | BS in BA: Information Systems & Business Analytics | 2021 – 2025

NCAA Division I Cross Country & Track, Dean’s List, America East Commissioner’s Honor Roll

WORK EXPERIENCE

Nassau Financial Group, Hartford, CT

June 2025 – August 2025

Data Engineer Intern (Solo Project Lead) | [View TPA Compliance Report](#)

- Developed a self-serve R Shiny analytics platform delivering risk-categorized insights across 7 activity streams to a 12-person compliance team, enabling data-driven prioritization of 120K+ insurance policies.
- Designed an activity-specific compliance scoring engine with configurable weighted checklists and pass thresholds across 8 insurance activity types, cutting manual TPA review time by ~50% via 12-month rolling quality trend analysis.

Sprague Operating Resources, Portsmouth, NH

June 2024 – August 2024

Business Intelligence & Data Intern

- Wrote complex SQL queries against commodity contract databases to extract and transform trading data; modeled results into a Power BI star schema with DAX-calculated measures to track energy position exposure across portfolios.

PROJECTS

Macroeconomic Anomaly Detection | [Live Demo](#)

January 2026 - March 2026

- Trained a PyTorch LSTM Autoencoder on FRED data (7 indicators, 12-month windows) to detect multivariate co-movement anomalies via reconstruction error; correctly flagged all 3 macro crises (2008 GFC, COVID-19, 2022 Fed hike cycle).
- Grounded multivariate design in VECM cointegration analysis; calibrated anomaly threshold at mean + 2 std on a pre-2007 baseline and deployed on Hugging Face Spaces with MLflow tracking and Llama-3.1-8B text-based explanations.

Airbnb Booking Prediction: Causal Drivers of Conversion | [View Report](#)

March 2026 - May 2026

- Built a CatBoost binary classifier on 374K US Airbnb listings extended to a 93k-row Paris panel with 45 features, 5-fold stratified CV, and Optuna-tuned hyperparameters across 40 Bayesian trials; achieved holdout AUC 0.966.
- Isolated host operating scale as an independent causal signal, where host type retained 0.110 SHAP after four price features; 15-year cohort analysis confirmed stable market composition since 2012 with no commercial recovery trajectory.

MLOps Loan Default Prediction | [View Report](#)

December 2025

- Built a financial risk prediction pipeline on 148K loan applications achieving 0.92 ROC-AUC; implemented PSI drift monitoring across 51 features to flag distribution shifts and surface emerging model risk post-deployment.
- Engineered a PyTorch DataLoader pipeline reducing memory usage 4,600×, enabling scalable processing of large financial datasets; trained Random Forest and deployed production model via MLflow and Docker with a live inference interface.

Financial News RAG Analyst | [Live Demo](#)

January 2026 - Present

- Built a production RAG system enabling queries across 500+ financial articles using LangChain, sentence-transformers, and ChromaDB; achieved 92% retrieval precision and <15ms latency, deployed on Hugging Face with Docker.
- Ran A/B tests on retrieval window size to select an optimal 100-article context window; integrated FinBERT sentiment scoring to surface sector-level insights, reducing research time from 30+ minutes to under 5 seconds.

TECHNICAL SKILLS

Core: Python (PyTorch, scikit-learn, pandas, LangChain), SQL, R; Random Forest, CatBoost, LightGBM, NLP, RAG, LLMs, Time Series (VECM, ARIMA), Feature Engineering, A/B testing, Causal Inference, Bayesian optimization

MLOps: Docker, MLflow, AWS (S3, Lambda), Hugging Face, FastAPI, Git/GitHub, Power BI, Tableau, DAX, LLMs, Agents